JAMES CHRISTOPHER HARTLEY

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PROFESSIONAL SUMMARY

Results-driven and highly motivated Research Engineer with a proven track record of leveraging cutting-edge technologies to spearhead innovative solutions within cross-functional teams. Passionate about delivering actionable insights through data-driven decision making to drive organisational objectives. Champion of mission-critical research initiatives, synergistically collaborating with stakeholders to enable dynamic capabilities. Committed to translating complex technical challenges into measurable value through proven expertise in emerging platforms and technologies.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, JavaScript

Research Tools: TensorFlow, PyTorch, scikit-learn, OpenCV

Data Technologies: SQL, PostgreSQL, Apache Spark, Hadoop

Cloud Platforms: AWS EC2, Google Cloud Platform, Azure

Development Tools: Git, Docker, Kubernetes, Jenkins

Methodologies: Machine Learning, Computer Vision, Data Analysis, Agile

Additional: REST APIs, Linux, Jupyter Notebooks, Pandas, NumPy

WORK EXPERIENCE

Research Engineer

TechVenture Labs

https://techventurelabs.io

Leeds, United Kingdom

June 2023 - Present

Developed and optimised machine learning models for computer vision applications, achieving 12% improvement in inference speed through architecture refinement and quantisation techniques.

Conducted experimental research on novel neural network architectures and documented findings through technical reports and peer presentations to cross-functional research teams.

Implemented data pipeline infrastructure using Python and Apache Spark to process 500GB+ of training datasets with automated validation and quality checks.

Collaborated with product engineering team to integrate research prototypes into production systems, ensuring scalability and maintaining model performance benchmarks.

Junior Research Engineer

DataSense Analytics Ltd

https://datasense-analytics.co.uk

Leeds, United Kingdom

January 2023 - May 2023

Engineered ETL pipelines using Python and SQL to process structured data from multiple sources, enabling faster data-driven insights for business intelligence teams.

Assisted senior researchers in designing and executing machine learning experiments for classification problems, managing experiment tracking and hyperparameter optimisation.

Developed RESTful APIs in Java to expose trained models for consumption by external applications and documented API specifications for integration teams.

Performed statistical analysis on experimental results and created visualisation dashboards using Matplotlib and Seaborn to communicate findings to non-technical stakeholders.

Graduate Research Placement

Innovation Research Centre

https://innovation-research-centre.net

Leeds, United Kingdom

July 2022 - December 2022

Built proof-of-concept prototypes for deep learning applications using TensorFlow and Keras, demonstrating feasibility of novel approaches for image segmentation tasks.

Participated in literature review and methodology workshops, consolidating research findings and contributing to technical documentation for project deliverables.

Conducted performance benchmarking tests across different hardware configurations to optimise computational resource allocation and identify cost-saving opportunities.

Supported lab infrastructure maintenance and managed version control systems, ensuring reproducibility of research workflows and code quality standards.

PUBLIC REPOSITORIES

GitHub: github.com/jchartley/ml-image-classification

A comprehensive PyTorch implementation of transfer learning models for multi-class image classification with data augmentation and evaluation metrics.

GitHub: github.com/jchartley/data-pipeline-framework

Python-based ETL framework for processing and transforming large-scale datasets with schema validation and error handling capabilities.

GitHub: github.com/jchartley/research-experiment-tracker

Experiment tracking and logging tool designed for machine learning research workflows with hyperparameter management and results visualisation.

GitHub: github.com/jchartley/sentiment-analysis-nlp

Natural language processing project using scikit-learn and spaCy for sentiment classification on social media text data.

EDUCATION

Bachelor of Science in Computer Science

University of Leeds

Leeds, United Kingdom

Graduated: June 2022

Grade: Upper Second Class Honours (2:1)

Relevant Modules: Machine Learning, Data Structures, Algorithms, Software Engineering, Artificial Intelligence, Database Systems

CERTIFICATIONS AND ADDITIONAL TRAINING

AWS Certified Cloud Practitioner (2023)

TensorFlow Developer Certificate (2023)

Python Programming Specialisation, Coursera (2022)